Static Shape of a Flexible Cable Hanging Between Two Pinned Points

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Abstract

A uniform, perfectly flexible, inextensible cable is pinned at two arbitrary points in a vertical plane and loaded only by its own weight. We derive, from a free-body balance of an infinitesimal element, the governing ordinary differential equation and integrate it in closed form. The equilibrium shape is a *catenary*, $y = y_0 + a\cosh\big((x - x_0)/a\big)$. We then solve the boundary-value problem posed by the two endpoints and the prescribed arc length, reducing the three matching conditions to a single transcendental equation for the catenary parameter a, give the existence criterion, and note the parabolic small-sag limit. A closing section records the analytical value of the result as a ground-truth benchmark for numerical cable models.

1 Statement and assumptions

A cable hangs in a vertical plane, fixed at two points, under its own weight. We seek its shape $y(x)$. The model rests on three assumptions.

(A1) **Perfectly flexible.** The cable carries no bending moment, so the internal force—the tension T—is everywhere *tangent* to the curve. This is the property that distinguishes a cable from a rod (a rod would add a bending term EI, raising the order of the governing equation; see Section 14).

(A2) **Inextensible and uniform.** The total arc length is a fixed constant L, and the weight per unit *arc length* is a constant $w = \lambda g$, with λ the linear mass density and g gravitational acceleration.

(A3) **Gravity only.** The sole distributed load is the vertical weight; the only other forces are the reactions at the two pins.

Coordinates: x horizontal, y vertical (positive up). Let s be arc length measured along the cable and $\theta(s)$ the angle the tangent makes with the horizontal, so that $\tan\theta = \mathrm{d}y/\mathrm{d}x$.

2 Free-body diagram of a cable element

Isolate an infinitesimal element of cable between arc lengths s and $s + \mathrm{d}s$. Three forces act on it:

- the tension $T(s)$ at the lower cut, directed tangentially "back-and-down";
- the tension $T(s + \mathrm{d}s)$ at the upper cut, directed tangentially "forward-and-up";
- the weight $w\,\mathrm{d}s$, directed straight down.

Resolve the tension into horizontal and vertical components,

$$H \equiv T\cos\theta, \qquad V \equiv T\sin\theta. \tag{1}$$

Static equilibrium of the element yields one scalar equation per direction:

$$\underbrace{\mathrm{d}(T\cos\theta) = 0,}_{\text{horizontal}} \qquad \underbrace{\mathrm{d}(T\sin\theta) = w\,\mathrm{d}s}_{\text{vertical}}. \tag{2}$$

3 The two conservation facts

Horizontal. The first relation in (2) integrates immediately:

$$\boxed{H = T\cos\theta = \text{constant}}. \tag{3}$$

This is the pivotal fact of the whole problem: because nothing loads the cable horizontally, the horizontal component of tension is identical at every point.

Vertical. The second relation gives $\mathrm{d}V/\mathrm{d}s = w$. Measure s from the *vertex*—the lowest point, where the tangent is horizontal, so that $\theta = 0$ and $V = 0$ there. Integrating from the vertex,

$$\boxed{V = T\sin\theta = w\,s}. \tag{4}$$

The vertical pull at any point equals the weight of cable hanging between that point and the vertex.

4 From forces to a slope equation

The slope of the curve is the ratio of the tension components,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\theta = \frac{T\sin\theta}{T\cos\theta} = \frac{V}{H} = \frac{w\,s}{H}. \tag{5}$$

Defining the *catenary parameter*

$$a \equiv \frac{H}{w} \qquad [\,\text{units of length}\,], \tag{6}$$

the slope equation takes the compact form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{s}{a}. \tag{7}$$

Remark. a is the single parameter that fixes the shape: it is the horizontal tension expressed in "weight-lengths." Large a (taut cable, large H) gives a flat curve; small a (slack cable) gives a deep sag.

5 Closing the equation with the arc-length relation

Equation (7) couples y to s, and s is itself defined by the curve via

$$\mathrm{d}s = \sqrt{1 + (y')^2}\,\mathrm{d}x, \qquad y' \equiv \frac{\mathrm{d}y}{\mathrm{d}x}. \tag{8}$$

Differentiating (7) with respect to x and substituting $\mathrm{d}s/\mathrm{d}x$,

$$\frac{\mathrm{d}y'}{\mathrm{d}x} = \frac{1}{a}\frac{\mathrm{d}s}{\mathrm{d}x} = \frac{1}{a}\sqrt{1 + (y')^2}. \tag{9}$$

This is the governing ODE: a first-order, separable equation for the slope $p \equiv y'$.

6 First integration: the slope

Separating variables in (9),

$$\frac{\mathrm{d}p}{\sqrt{1+p^2}} = \frac{\mathrm{d}x}{a}. \tag{10}$$

The left-hand side integrates to $\operatorname{arsinh} p$, since $\frac{\mathrm{d}}{\mathrm{d}p}\operatorname{arsinh} p = 1/\sqrt{1+p^2}$, giving

$$\operatorname{arsinh}(p) = \frac{x}{a} + C_1. \tag{11}$$

Placing $x = 0$ at the vertex, where $p = y' = 0$, fixes $C_1 = \operatorname{arsinh}(0) = 0$. Inverting,

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh\frac{x}{a}}. \tag{12}$$

7 Second integration: the shape

Integrating (12),

$$y = \int \sinh\frac{x}{a}\,\mathrm{d}x = a\cosh\frac{x}{a} + C_2. \tag{13}$$

The constant C_2 merely places the curve vertically. Writing it together with a horizontal shift gives the *catenary*:

$$\boxed{y(x) = y_0 + a\cosh\frac{x - x_0}{a}}, \tag{14}$$

whose vertex sits at $\bigl(x_0,\, y_0 + a\bigr)$. The shape is a hyperbolic cosine—*not* a parabola (Section 13).

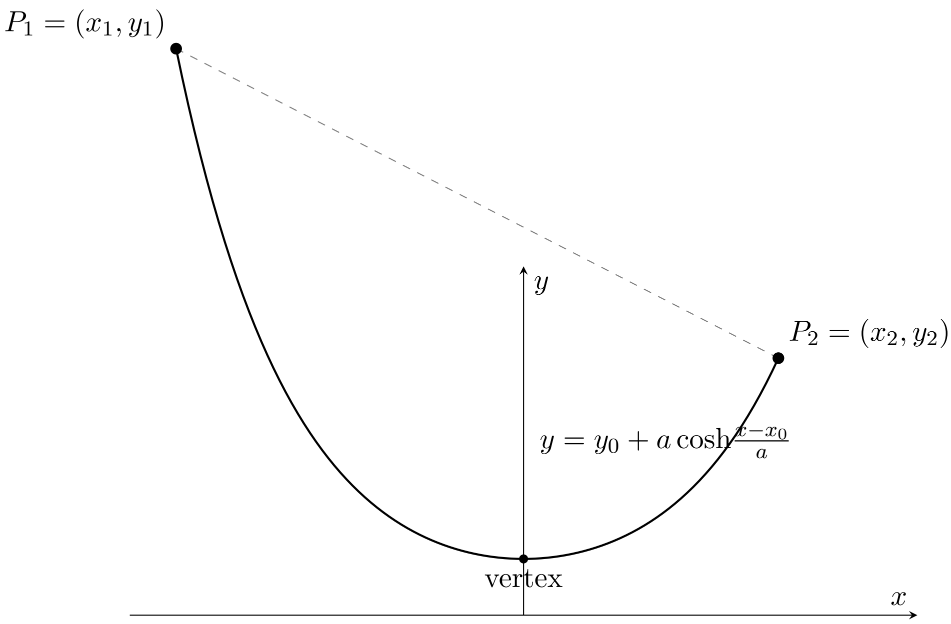

Figure 1: A catenary pinned at two points of unequal height. The dashed line is the chord of length $\sqrt{h^2+v^2}$; the cable's arc length between the pins is $L > \sqrt{h^2+v^2}$. At the vertex the tension is purely horizontal and equal to $H = wa$.

8 Arc length and a built-in consistency check

Using (12) and the identity $1 + \sinh^2 = \cosh^2$,

$$s(x) = \int_0^x \sqrt{1 + \sinh^2 \tfrac{x'}{a}}\, \mathrm{d}x' = \int_0^x \cosh \tfrac{x'}{a}\, \mathrm{d}x' = a \sinh \frac{x}{a}. \tag{15}$$

Substituting back into (7) returns $y' = s/a = \sinh(x/a)$, in agreement with (12): the derivation closes on itself.

Remark (Tension distribution). From (3) and $\cos\theta = 1/\sqrt{1 + \sinh^2} = 1/\cosh\frac{x-x_0}{a}$, the tension is

$$T = H \cosh \frac{x - x_0}{a} = w\,(y - y_0).$$

Tension is proportional to height above the directrix $y = y_0$. The minimum, $H = wa$, occurs at the vertex; the maximum occurs at the higher support. This is the relation to consult when checking whether a cable, or a discretized joint chain, will exceed its breaking load.

9 The boundary-value problem

The catenary (14) carries three unknowns: a (shape), x_0 (vertex abscissa), and y_0 (vertical offset). The data supply exactly three conditions—the two pins and the length:

$$\text{(i) } y(x_1) = y_1, \qquad \text{(ii) } y(x_2) = y_2, \qquad \text{(iii) } s(x_2) - s(x_1) = L. \tag{16}$$

Introduce the dimensionless endpoint arguments

$$A \equiv \frac{x_1 - x_0}{a}, \qquad B \equiv \frac{x_2 - x_0}{a}, \tag{17}$$

so that the three conditions read

$$y_1 = y_0 + a \cosh A, \qquad y_2 = y_0 + a \cosh B, \qquad L = a\,(\sinh B - \sinh A). \tag{18}$$

10 Reduction to a single transcendental equation

Let $h \equiv x_2 - x_1$ be the horizontal span and $v \equiv y_2 - y_1$ the vertical rise. Subtracting (i) from (ii) eliminates y_0:

$$v = a\,(\cosh B - \cosh A), \qquad L = a\,(\sinh B - \sinh A). \tag{19}$$

Apply the sum-to-product identities

$$\cosh B - \cosh A = 2 \sinh \tfrac{B+A}{2} \sinh \tfrac{B-A}{2}, \tag{20}$$

$$\sinh B - \sinh A = 2 \cosh \tfrac{B+A}{2} \sinh \tfrac{B-A}{2}. \tag{21}$$

Since $B - A = (x_2 - x_1)/a = h/a$, we have $\frac{B-A}{2} = \frac{h}{2a}$. Writing $m \equiv \frac{A+B}{2}$,

$$v = 2a\, \sinh m\, \sinh \frac{h}{2a}, \qquad L = 2a\, \cosh m\, \sinh \frac{h}{2a}. \tag{22}$$

Combine the two equations in (22) in two ways. *Dividing* cancels the common factor $2a \sinh \frac{h}{2a}$:

$$\frac{v}{L} = \tanh m \quad \Longrightarrow \quad m = \operatorname{artanh} \frac{v}{L}. \tag{23}$$

Squaring and subtracting, with $\cosh^2 m - \sinh^2 m = 1$,

$$L^2 - v^2 = \left(2a \sinh \tfrac{h}{2a}\right)^2, \tag{24}$$

which yields the master equation—a single transcendental equation in the lone unknown a:

$$\boxed{\sqrt{L^2 - v^2} = 2a\, \sinh \frac{h}{2a}}. \tag{25}$$

11 Solution recipe

1. **Solve** (25) **for** a. The right-hand side $f(a) = 2a\sinh\frac{h}{2a}$ is monotone (Section 12), so Newton's method or bisection converges quickly. A serviceable initial guess follows from the small-sag relation $f(a) \approx h\big(1+\frac{h^2}{24a^2}\big)$, i.e. $a_0 \approx h\big/\sqrt{24\,(\sqrt{L^2-v^2}/h-1)}$.

2. **Vertex abscissa** x_0 from (23). Since $m = \frac{A+B}{2} = \dfrac{x_1+x_2-2x_0}{2a}$,

$$x_0 = \frac{x_1+x_2}{2} - a\ \mathrm{artanh}\,\frac{v}{L}. \tag{26}$$

3. **Vertical offset** y_0 from condition (i):

$$y_0 = y_1 - a\cosh\frac{x_1-x_0}{a}. \tag{27}$$

4. **Assemble** the shape:

$$y(x) = y_0 + a\cosh\frac{x-x_0}{a}. \tag{28}$$

12 Existence and the taut limit

Consider $f(a) = 2a\sinh\frac{h}{2a}$. As $a \to 0^+$ the hyperbolic sine dominates and $f \to \infty$; as $a \to \infty$, $\sinh\varepsilon \approx \varepsilon$ gives $f \to 2a\cdot\frac{h}{2a} = h$. The function decreases monotonically from ∞ to h. Hence (25) has a unique positive root if and only if

$$\sqrt{L^2-v^2} > h \iff L > \sqrt{h^2+v^2}. \tag{29}$$

The cable must be longer than the chord. When L equals the chord, $a \to \infty$ and the catenary degenerates to a straight, taut line; L less than the chord is impossible for an inextensible cable—precisely the expected physical constraint.

13 Why it is a cosh and not a parabola

A suspension-bridge main cable *is* parabolic, because there the load is uniform per unit *horizontal* distance (the deck), giving $V = w_x x$ and $y'' =$ const. Here the load is uniform per unit *arc length* (the cable's own weight), which produces the cosh. The parabola is the small-sag limit of the catenary: for $|x-x_0| \ll a$,

$$a\cosh\frac{x-x_0}{a} \approx a + \frac{(x-x_0)^2}{2a}, \tag{30}$$

the leading term of the Taylor expansion. A shallow catenary therefore looks parabolic, but the exact self-weight solution is the hyperbolic cosine.

14 The result as a numerical benchmark

Equation (14) with the constants of the solution recipe is the analytical ground truth for a static hanging cable, and as such is the canonical check for numerical rod and cable models:

- A **position-based (XPBD) inextensible rod** should relax to this catenary. Residual error indicates insufficient constraint stiffness or too few solver iterations, not a modelling defect.

- A **capsule chain with free angular joints** matches the catenary; any non-zero joint stiffness appears as a systematic departure from cosh that this solution quantifies exactly.

- A **volumetric FEM cable** will *not* match it, because it absorbs axial load as stretch. Its rest shape is the *elastic catenary*, in which the arc-length–tension relation acquires a stretching term proportional to $1/(EA)$; under identical endpoints it sags more than the inextensible solution. That is the appropriate ground truth for the two-way FEM case and is a natural next derivation.

A final caveat on scope: the present result assumes zero bending stiffness. A rod with non-negligible EI is no longer a catenary but a gravito-elastica, governed by a fourth-order ODE with a boundary layer near each clamped end. For a cable, $EI \approx 0$ and the catenary is exact.